

Proposal Guide

Your MS1 proposal: max. 2-page PDF, uploaded to ILIAS "MS1 Submission" with your repo URL, copy committed as docs/proposal.pdf. Use these four headings; each is graded as one criterion.

A checklist for your own document, not a form to fill in. Deadlines and weights: see the project slides.

YOUR DOCUMENT

Project Proposal — <project title>

Author · Repository (public GitHub URL of the project repo)

1 Problem statement

- ☐ What do you predict, and for whom?
 - ☐ Exact horizon ("60 minutes ahead", not "soon") and scope: which city, stations, assets, period?
 - ☐ Success criterion with a number and a baseline.
- STRONG** "Predict per-station bike availability 60 min ahead, $MAE \leq 2$, beating persistence."
- WEAK** "Analyse bike-sharing data."

2 Originality & motivation

- ☐ Why this problem, and why you?
- ☐ Checked against past HSLU projects (ml_ops-lab.ch) and the KTH ID2223 project lists?
- ☐ One sentence on what makes yours different. Crypto direction, air quality and news classification have been done repeatedly: not banned, but an undifferentiated rerun scores low.

3 Data source & features

- ☐ Live source and provider; API or scraping? If scraping: the terms of service allow it, and you name a fallback source.
- ☐ Update frequency. Dynamic data is required; a static CSV does not qualify.
- ☐ How much data exists today and how fast it grows. Say so if history only starts when your poller starts.
- ☐ The label: what it is and where it comes from. It must not be mechanically derivable from your features.
- ☐ Rare positive class: how you handle and evaluate it.
- ☐ Named features, not "various features".
- ☐ Leakage: out-of-time split, and where future information could leak into a feature.

4 System design

- ☐ FTI architecture diagram embedded as an image (a draft is fine; not referenced, not ASCII).
- ☐ Tech stack: feature store, experiment tracking, orchestration, serving, with one line why each.
- ☐ Stretch layers (heavy orchestration, IaC, managed cloud extras, extra monitoring) explicitly marked optional; core FTI first.
- ☐ Confirm the repository is public.

FORMAT

- max. 2 pages
- PDF
- ILIAS "MS1 Submission" + repo URL
- copy committed as docs/proposal.pdf

COMMON MISTAKES

- no baseline
- diagram missing or not embedded
- label derived by a rule from the inputs
- recycled topic without a twist
- stack too ambitious for the core